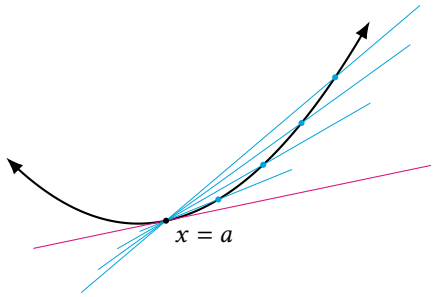
Lecture 2: A Review of Derivatives

13 August 2025

Recall the tangent problem: suppose we wished to compute the slope of the tangent line to the curve $y = f(x)$ at the point where $x = a$.



By drawing lines to points that get closer and closer to $x = a$, we can thus approximate the slope of the tangent line with

$$\text{slope of the tangent line} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

This gives us the definition of the **derivative** of $f(x)$ at $x = a$. If this derivative exists, we say that $f(x)$ is **differentiable** at $x = a$. We may express the derivative of f as a function of x:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

From this, we are able to compute **higher-order derivatives**: for instance, differentiating f', we get the **second derivative**

$$f''(x) = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2},$$

Differentiating again yields the **third derivative**

$$f'''(x) = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3y}{dx^3},$$

and so on. More generally, we can compute the ***n*th derivative**

$$f^{(n)}(x) = \frac{d^ny}{dx^n}.$$

Curve Sketching

The derivative of a function $y = f(x)$ tells us how the function is changing: in particular, it can tell us where $f(x)$ is increasing or decreasing.

- If $f'(x) > 0$ on an interval, then $f(x)$ is *increasing* on that interval.
- If $f'(x) < 0$ on an interval, then f is *decreasing* on that interval.

The points at which $f'(x) = 0$ are called **stationary points** (or **critical points**): at these points, the slope of the tangent line is zero, and so the graph may attain extreme values at these points.

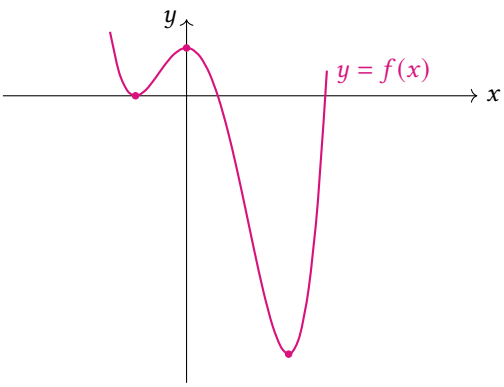
Example. Consider the graph of the function

$$f(x) = 3x^4 - 4x^3 - 12x^2 + 5.$$

Its derivative is given by

$$f'(x) = 12x^3 - 12x^2 - 24x = 12(x+1)x(x-2).$$

The graph thus has stationary points at the points where $f'(x) = 0$: $x = -1$ at which $f(-1) = 0$, $x = 0$ at which $f(0) = 5$, and $x = 2$ at which $f(1) = -27$. By plotting these points, we can sketch the graph of $y = f(x)$.



Note that the *sign* of the derivative can also tell us whether the function is increasing or decreasing outside of these points.

x	$x+1$	x	$x-2$	$f'(x)$	$f(x)$
$x < -1$	$-$	$-$	$-$	$-$	decreasing ↘
$-1 < x < 0$	$+$	$-$	$-$	$+$	increasing ↗
$0 < x < 2$	$+$	$+$	$-$	$-$	decreasing ↘
$x > 2$	$+$	$+$	$+$	$+$	increasing ↗

We can classify the stationary points of our graph: we say that $f(x)$ has a **local maximum** at $(0, 5)$, and two **local minimum** points at $(-1, 0)$ and $(2, -27)$. Furthermore, observe that the point $(2, -27)$ is a **global minimum** as well, since $f(2) = -27$ is less than (or equal to) $f(x)$ for all possible values of x.

We summarize a guide to curve sketching as follows:*

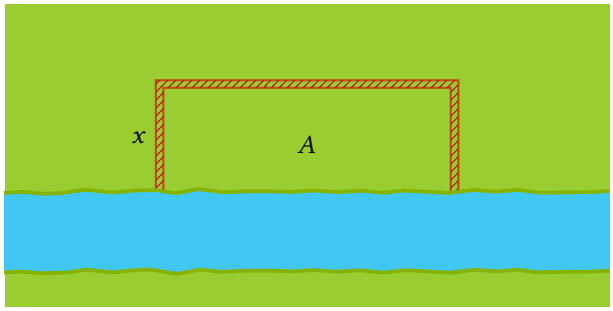
1. determine the **domain** of $f(x)$: the values of x for which $f(x)$ is defined,
2. find the **stationary points** of $f(x)$ by determining the values of c at which $f'(c) = 0$,
3. determine the **intervals of increase or decrease** by using the first derivative: if $f'(x) > 0$ then f is increasing, and if $f'(x) < 0$ then f is decreasing,
4. classify **local extrema**: if f' goes from positive to negative at a critical value c, then $f(c)$ is a local maximum, and if f' goes from negative to positive at c, then $f(c)$ is a local minimum.

* For a more precise sketch, you can determine the concavity of different intervals by computing the second derivative (see *Extra Notes on Derivatives*).

Optimization

The derivative of a function allows us to determine when a function attains a maximum or a minimum. This is particularly helpful when we wish solve problems that ask us to find extreme values in practical application: we might be interested in maximizing areas, volumes, profits, or minimize distances, costs, times, etc.

Example. You wish to fence off a rectangular field that borders a river. If you have 2400 feet of fencing, what are the dimensions of the field that has the largest possible area?



Let x be the width of the fenced area. Then, the length of the rectangle must be $2400 - 2x$, and we wish to maximize the area

$$A(x) = x(2400 - 2x) = 2400x - 2x^2.$$

To determine the maximum point, we consider the first derivative $A'(x)$ and set it to zero:

$$A'(x) = 2400 - 4x = 0 \quad \Longrightarrow \quad x = 600.$$

Thus, $A(x)$ must have a maximum value when $x = 600$. We can check that this indeed corresponds to a maximum value:

- when $x < 600$, $A'(x) > 0$ and $A(x)$ is increasing, and
- when $x > 600$, $A'(x) < 0$ and $A(x)$ is decreasing.

We conclude that the fenced area must width $x = 600$ feet and length $2400 - 2x = 1200$ feet, and the maximal area is

$$A(600) = 600 \times 1200 = 720000 \text{ square feet.}$$

Exercise. Suppose you wish to make a cylindrical can that can hold, at most, 1 cubic meter of oil. Find the dimensions that will *minimize* the cost of the metal to manufacture the can.

Try this problem out! You should find that the cost is minimized when the can's radius is $r = 1/\sqrt[3]{2\pi}$ and its height is $h = \sqrt[3]{4/\pi}$.

l'Hôpital's Rule

One notable application of derivatives is in computing difficult limits. Limits of the form

$$\lim_{x \to a} \frac{f(x)}{g(x)}, \text{ where } \lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0,$$

or

$$\lim_{x \to a} \frac{f(x)}{g(x)}, \text{ where } \lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \pm\infty,$$

are called **indeterminate forms**. L'Hôpital's Rule, named after the French mathematician Guillaume Françis Antoine, Marquis de l'Hôpital (1661–1704), gives us a way of evaluating such limits.

Theorem (l'Hôpital's Rule)**.** *Let f and g be differentiable functions, with $g'(x) = 0$ on an open interval that contains a (except possibly at a). Suppose that*

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = 0,$$

or that

$$\lim_{x \to a} f(x) = \lim_{x \to a} g(x) = \pm\infty.$$

Then,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)},$$

provided that the limit on the right side exists (or is ∞ or $-\infty$).

Note that it is important to verify that the given conditions regarding the limits of f and g are satisfied before using l'Hôpital's rule.

Remark. l'Hôpital's rule works in many other situations: in particular, you can use the rule to evaluate *one-sided limits*, and you can also use the rule to evaluate *limits at infinity*.

Extra Notes on Derivatives

Differentiation Rules. We have a few rules on derivatives which we often assume and must thus be familiar with. Let f and g be differentiable functions. Then,

1. *Linearity:* $(cf)' = cf'$ for all constants c and $(f+g)' = f' + g'$
2. *The Product Rule:* $(fg)' = f'g + fg'$
3. *The Quotient Rule:* $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$, given $g \neq 0$
4. *The Chain Rule:* $(f(g(x)))' = f'(g(x)) \cdot g'(x)$

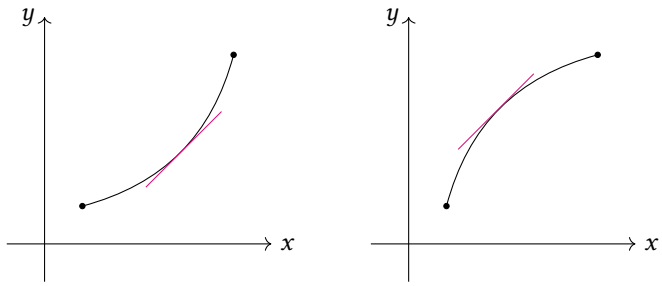
We have the following standard formulae:

$$\frac{d}{dx}c = 0, \quad \frac{d}{dx}x^n = nx^{n-1}, \quad \frac{d}{dx}e^x = e^x, \quad \frac{d}{dx}\ln x = \frac{1}{x},$$
$$\frac{d}{dx}\sin x = \cos x, \quad \frac{d}{dx}\cos x = -\sin x, \quad \frac{d}{dx}\tan x = \frac{1}{\cos^2 x}.$$

We also have the following derivatives of inverse functions, which can be computed by implicit differentiation

$$\frac{d}{dx}\sin^{-1} x = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\cos^{-1} x = -\frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}\tan^{-1} x = \frac{1}{1+x^2}.$$

Second Derivatives and Concavity. Consider two graphs with the same end points.



Observe that both functions are increasing (and would have a positive first derivative), but they look different: we define the **concavity** of a function to distinguish between them.

- If the graph of a function lies *above* all of its tangent lines on an interval, as shown on the left, then the graph is said to be **concave up**.
- If the graph of a function lies *below* all of its tangent lines on an interval, as shown on the right, then it is said to be **concave down**.

Observe that if a graph is concave up, the slopes $f'(x)$ of the tangent lines are increasing, whereas if a graph is concave down, the slopes of the tangent lines are decreasing. We can thus use the second derivative $f''(x)$ to determine the concavity of a graph on an interval.

Suppose that $f(x)$ is twice differentiable.

- If $f''(x) > 0$ on an interval, then the graph of $f(x)$ is *concave up* on that interval.
- If $f''(x) < 0$ on an interval, then the graph of $f(x)$ is *concave down* on that interval.